Estimating Energy Consumption

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1 Receiver Energy

The energy consumption of the receiver can be estimated from the number of real multiplications N_M and real additions N_A it peforms to process each symbol. The true cost of these operations depends on the hardware implementation. Here the reference values shown in table 1 are used to allow for comparison between different DSP algorithms. An accurate estimate of their actual energy consumption would require analysis of their ASIC implementation for a modern process node.

Table 1: Energy cost of arithmetic operations for different bit widths [1].

n (bits)	Add (pJ)	Multiply (pJ)
8	0.03	0.2
32	0.1	3.1

1.1 Dependence on Word Length

A straightforward adder implementation requires n bit-wise additions when operating on words of length n. Hence fitting a linear relationship through the origin to table 1 gives an estimate for the energy of an n-bit addition $E_A(n) = 3.16n$ fJ. Multiplication requires the accumulation of n partial products of length n, so scales with n^2. Now fitting a quadratic through the origin to table 1 gives the estimated energy of an n-bit multiplication $E_M(n) = 3.03n^2$ fJ.

1.2 Oversampling

Timing recovery and fractionally-spaced equalisation require the sampling rate f to be higher than the symbol rate R_s. For DSP blocks that operate on the oversampled signal, the number of multiplications and additions can first be calculated per sample then scaled by the oversampling rate $\frac{f}{R_s}$.

1.3 Total Energy and Power

The energy per symbol for the entire receiver is given by

$$E_s = \sum_{k=1}^{K} N_A(k)E_A(n_k) + N_M(k)E_M(n_k), \tag{1}$$

where the receiver is decomposed into K blocks that use different word lengths n_k according to their required precision. For N symbols per polarisation, the total energy

$$E_{\mathrm{rx}} = 2NE_s \tag{2}$$

and the total power

$$P_{\mathrm{rx}} = 2R_sE_s. \tag{3}$$

2 Transmitter Energy

2.1 Minimum Required SNR

It is not sufficient to only compare the energy consumption of receiver implementations if they differ in performance. The CPON standard specifies a maxium bit-error rate (BER) of 10^{-2} that can be allowed for the forward error-correction (FEC) scheme to work. Each receiver implementation will have a minimum required SNR to achieve the FEC limit. This FEC SNR determines, along with the noise present in the optical network, the minimum power at the transmitter.

2.2 Noise

The relationship between received SNR and transmitted power depends on the sources of noise in the network. Optical access networks can often be

References

[1] Mark Horowitz. "Computing's Energy Problem (and what we can do about it)". In: *2014 IEEE International Solid-State Circuits Conference (ISSCC) Digest of Technical Papers*. 2014, pp. 10–14. DOI: 10.1109/ISSCC.2014.6757323.